

Algebra 1
Unit 6
Lesson 1 Homework

Name Answer Key
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Jackson made a mistake when solving the inequality $\frac{-8x+20}{4} - 10 \geq 7$.
For questions 1- 4, use Jackson's work below.

Given Inequality	$\frac{-8x + 20}{4} - 10 \geq 7$
Step 1	$-8x + 20 - 10 \geq 28$
Step 2	$-8x + 10 \geq 28$
Step 3	$-8x \geq 18$
Step 4	$x \leq -\frac{18}{8}$

1. Identify the step in which Jackson made an error. Write a note to Jackson explaining why that step was incorrect.

The error was made in Step 1. Before you can multiply by 4, the -10 must be added to both sides.

2. Solve the inequality $\frac{-8x+20}{4} - 10 \geq 7$.

$$\frac{-8x+20}{4} \geq 17$$

$$4 \cdot \frac{-8x+20}{4} \geq 17 \cdot 4$$

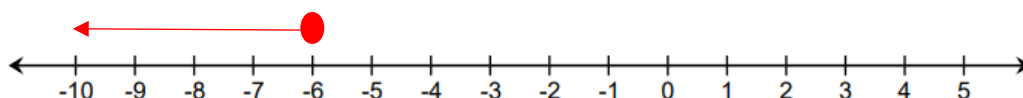
$$\begin{array}{r} -8x + 20 \geq 68 \\ -20 \quad -20 \end{array}$$

$$\frac{-8x}{-8} \geq \frac{48}{-8} \quad x \leq -6$$

3. Write the solution in set-builder notation.

$$\{x \in \mathbb{R} : x \leq -6\}$$

4. Graph the solution set.



Use the inequality $0.05x + 1.13 > -0.24x + 14.18$ to complete questions 5-7.

5. Solve the inequality.
$$\begin{array}{rcl} 0.05x + 1.13 & > & -0.24x + 14.18 \\ +0.24x & & + 0.24x \end{array}$$

$$\begin{array}{rcl} 0.29x + 1.13 & > & 14.18 \\ -1.13 & -1.13 & \end{array}$$

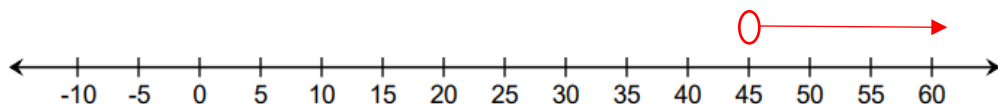
$$\frac{0.29x}{0.29} > \frac{13.05}{0.29}$$

$$x > 45$$

6. Write the solution in set-builder notation.

$$\{x \in \mathbb{R} : x > 45\}$$

7. Graph the solution set.



Use the inequality $-\frac{3}{2}y + \frac{2}{3} \leq -\frac{1}{3}(y + 12)$ to complete questions 8-10.

8. Solve the inequality.
$$\begin{array}{rcl} -\frac{3}{2}y + \frac{2}{3} & \leq & -\frac{1}{3}y - 4 \\ +\frac{1}{3}y & & +\frac{1}{3}y \end{array}$$

$$\begin{array}{rcl} -\frac{7}{6}y + \frac{2}{3} & \leq & -4 \\ -\frac{2}{3} & -\frac{2}{3} & \end{array}$$

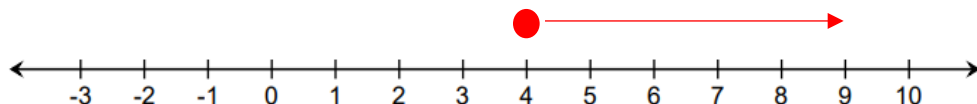
$$-\frac{6}{7} \cdot -\frac{7}{6}y \leq -\frac{14}{3} \cdot -\frac{6}{7}$$

$$y \geq 4$$

9. Write the solution in set-builder notation.

$$\{y \in \mathbb{R} : y \geq 4\}$$

10. Graph the solution set.



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Lesson 2 Homework

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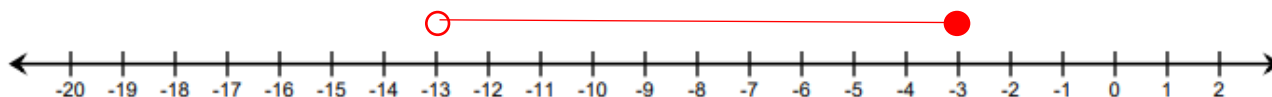
Use the following conjunctive inequality to answer questions 1-3.

$$5v + 10 \leq -4v - 17 < 9 - 2v$$

Rewrite the compound inequality as two single inequalities and solve each one.

1. $5v + 10 \leq -4v - 17$ $+4v \quad +4v$ $9v + 10 \leq -17$ $-10 \quad -10$ $\frac{9v}{9} \leq \frac{-27}{9}$ $v \leq -3$	2. $-4v - 17 < 9 - 2v$ $+2v \quad +2v$ $-2v - 17 < 9$ $+17 \quad +17$ $\frac{-2v}{2} < \frac{26}{-2}$ $v > -13$
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3. Graph the solution set of the compound inequality.



Kali just started a new sales floor job to save for college. She earns \$15.75 per hour plus a flat fee of \$50 per week. She wants to earn between \$200 and \$400 per week. The following inequality represents her earning potential:

$$200 \leq 15.75x + 50 \leq 400$$

4. Solve the inequality.

$$15.75x + 50 \geq 200$$

$$-50 \quad -50$$

$$\frac{15.75x}{15.75} \geq \frac{150}{15.75}$$

$$x \geq 9\frac{11}{21}$$

$$15.75x + 50 \leq 400$$

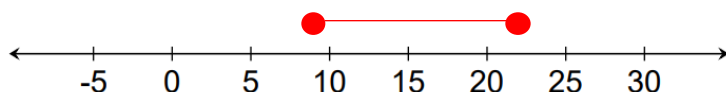
$$-50 \quad -50$$

$$\frac{15.75x}{15.75} \leq \frac{350}{15.75}$$

$$x \leq 22\frac{2}{9}$$

$$9\frac{11}{21} \leq x \leq 22\frac{2}{9}$$

5. Graph the solution on the number line. Approximate values as necessary.



6. Complete the statement: Kali can work between $9\frac{11}{21}$ and $22\frac{2}{9}$ hours each week.
 $\approx 9.5 \quad \approx 22.2$

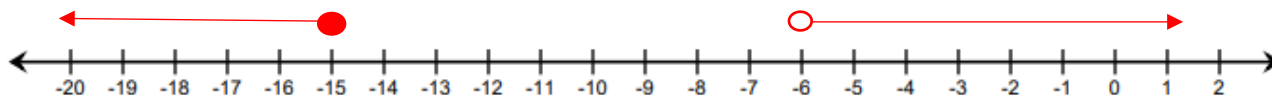
Use the disjunctive inequality shown below to answer questions 7-10:

$$-\frac{15}{4}\left(2x - \frac{2}{3}\right) < 64 + \frac{11}{4}x \quad \text{or} \quad \frac{35x - 11}{4} \leq -134$$

Solve each component of the compound inequality.

7. $-\frac{15}{4}\left(2x - \frac{2}{3}\right) < 64 + \frac{11}{4}x$ $\begin{array}{r} -7.5x + 2.5 < 64 + 2.75x \\ -2.75x \qquad \qquad -2.75x \\ \hline -10.25x + 2.5 < 64 \\ -2.5 \quad -2.5 \\ \hline \frac{-10.25x}{-10.25} < \frac{61.5}{-10.25} \\ x > -6 \end{array}$	8. $\frac{35x-11}{4} \leq -134$ $\begin{array}{r} 4 \cdot \frac{35x-11}{4} \leq -134 \cdot 4 \\ \hline 35x - 11 \leq -536 \\ +11 \quad +11 \\ \hline \frac{35x}{35} \leq \frac{-525}{35} \\ x \leq -15 \end{array}$
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9. Graph the solution sets of both inequalities on the number line.



10. Explain which of the following values is not a solution to the inequality.

-20	-10	0
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-10 is not included because neither inequality includes the solution set.

Paolo is saving for a trip to the Bahamas with his friends. He has estimated that his travel expenses will cost between \$2,000 and \$3,000 total. He has \$250 in savings, and he earns \$15.25 per hour at his job. He wants to know how many hours he must work in order to afford the trip.

1. Write an inequality to represent the situation.

$$2,000 \leq 250 + 15.25x \leq 3,000$$

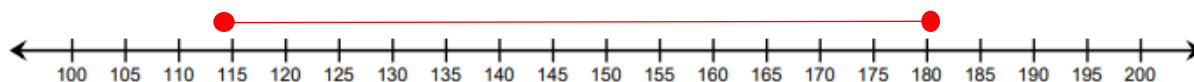
2. Solve the inequality.

$$\begin{array}{rcl} 2,000 & \leq & 250 + 15.25x \leq 3,000 \\ -250 & -250 & -250 \end{array}$$

$$\frac{1750}{15.25} \leq \frac{15.25x}{15.25} \leq \frac{2750}{15.25}$$

$$114.75 \leq x \leq 180.33$$

3. Graph the solution.



4. Check any that apply.

- ☒ Solutions must be non-negative.
☐ Solutions must be integers.

5. Explain your selection(s) from question 4.

Paolo can work fractions of hours but can't work negative hours.

Celine just started her own business making cake pops. She sells the cake pops in boxes, charging \$20.00 per box. It costs her \$65.00 to buy the supplies to make the cake pops. She wants to calculate how many boxes she must make to earn a profit of at least \$225.00.

6. Write an inequality to represent the situation.

$$20x - 65 \geq 255$$

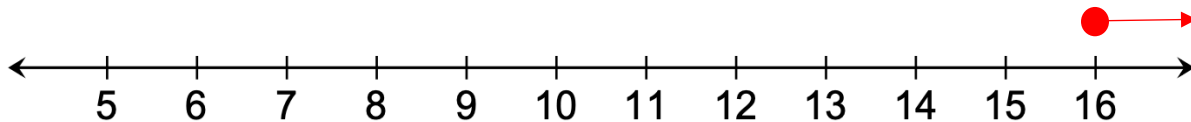
7. Solve the inequality.

$$\begin{array}{r} 20x - 65 \geq 255 \\ +65 \quad +65 \end{array}$$

$$\frac{20x}{20} \geq \frac{320}{20}$$

$$x \geq 16$$

8. Graph the solution.



9. Check any that apply.

☐ Solutions must be non-negative.

☐ Solutions must be integers.

10. Explain your selection(s) from question 9.

Celine can't sell negative boxes and she can't sell fractional parts of boxes.

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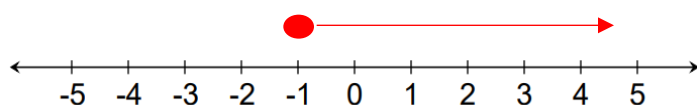
Consider the inequality $-2x + 5 \leq 7$.

1. Solve the inequality. $-2x + 5 \leq 7$
 $-5 -5$

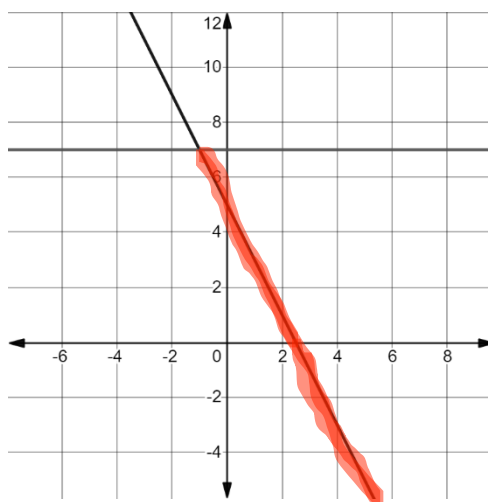
$$\frac{-2x}{-2} \leq \frac{2}{-2}$$

$$x \geq -1$$

2. Graph the solution set.



3. The graphs of $y = -2x + 5$ and $y = 7$ are shown on the coordinate plane.

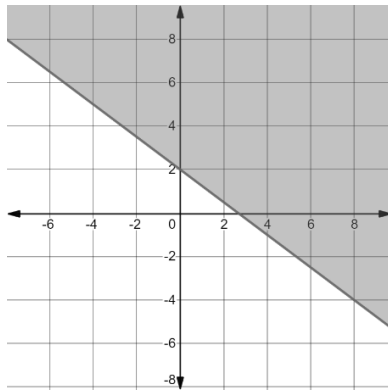


Highlight the part of the graph of $y = -2x + 5$, where the y values are less than or equal to 7.

4. Describe the relationship between the solution to the inequality $-2x + 5 \leq 7$ and the values of x for which the graph of $y = -2x + 5$ is less than or equal to 7.

When $x \geq -1$, the solution to $-2x + 5 \leq 7$, the y -values of $y = -2x + 5$ are ≤ 7 .

The graph of the two-variable inequality $y \geq -\frac{3}{4}x + 2$ is shown.



5. The slanted solid line is called the **boundary line** of the inequality. Write the equation of the boundary line in slope-intercept form.

$$y = -\frac{3}{4}x + 2$$

6. How does the equation of the boundary line relate to the inequality?

They have the same slope and y-intercept & are the same side from the = & ≥.

7. Describe what part(s) of the graph represent the solution set of the inequality $y \geq -\frac{3}{4}x + 2$.

The shaded parts contains all values that represent solutions.

8. Which part of the graph of the solution set of $y \geq -\frac{3}{4}x + 2$ corresponds to each mathematical statement?

Mathematical Statement	Graphical Representation
$y = -3/4x + 2$	☒ boundary line ☐ shaded region
$y > -3/4x + 2$	☐ boundary line ☒ shaded region

Complete the statements.

9. If the points on the line are included in the solution set, the boundary line

is ☒ dotted.
☒ solid.

10. If the points on the line are not included in the solution set, the boundary

line is ☒ dotted.
☐ solid.

Algebra 1
Unit 6
Lesson 5 Homework

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The linear inequality $3x - 4y < 12$ is written in standard form.

1. Complete the statement.

The boundary line of this inequality is

☒ dotted
☐ solid

because the points on the

boundary line are

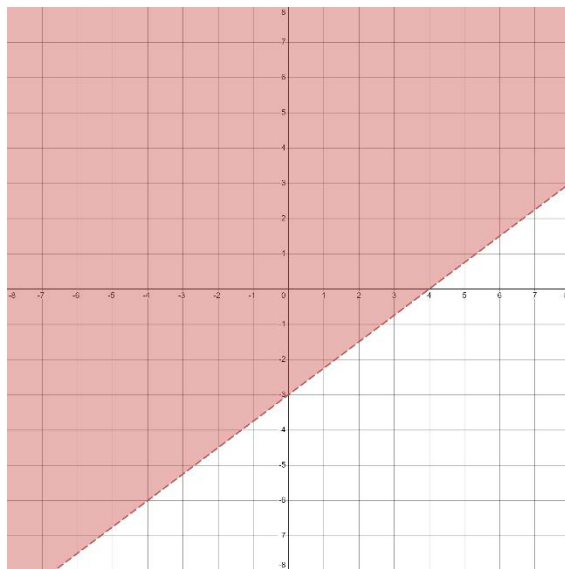
☐ included in
☒ excluded in

the set.

2. Write the equation of the boundary line of this inequality in standard form.

$$3x - 4y = 12$$

3. Graph the boundary line on the coordinate grid.



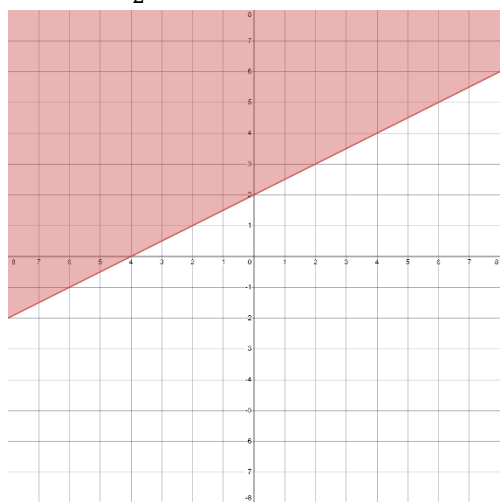
4. Explain how to determine which side of the boundary line to shade.

Choose a point, plug it in. If the inequality is true, shade in the same side as the point. If it's not true, shade in the opposite side.

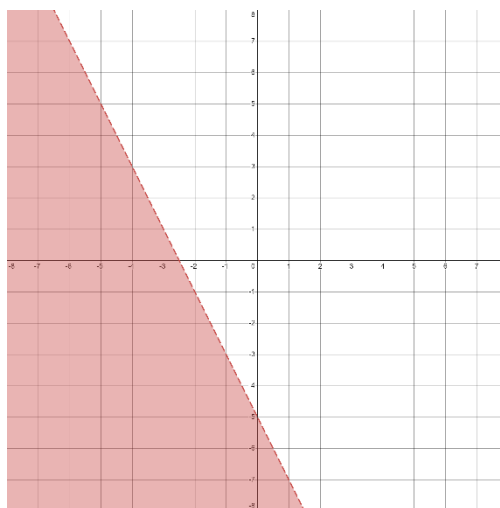
5. Shade the region on the graph that represents the solution set.

For each problem below, graph the given inequality.

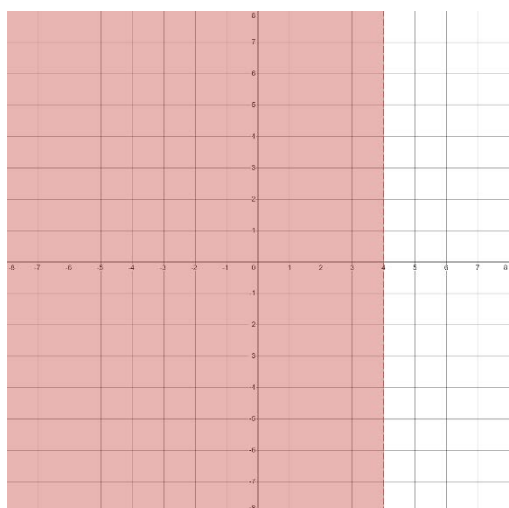
6. $y \geq \frac{1}{2}x + 2$



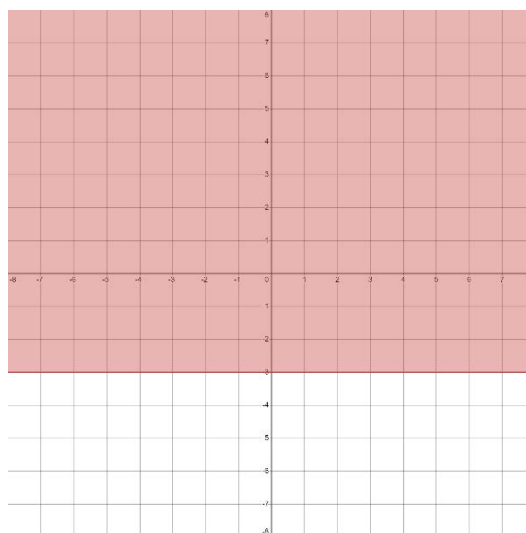
7. $y < -2x - 5$



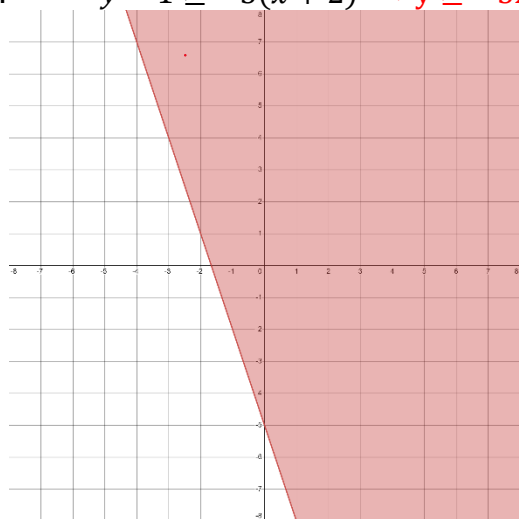
8. $x < 4$



9. $y \geq -3$



10. $y - 1 \geq -3(x + 2) \rightarrow y \geq -3x - 5$



Algebra 1

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Molly is buying lunchmeat and cheese from the Publix deli. Lunchmeat is \$5 per pound and cheese is \$4 per pound. Molly's mother gave her \$30 to spend. To determine the combinations of meat and cheese that she can afford to purchase, Molly wrote the inequality $5x + 4y \leq 30$.

1. Define the variables Molly used based on the context.

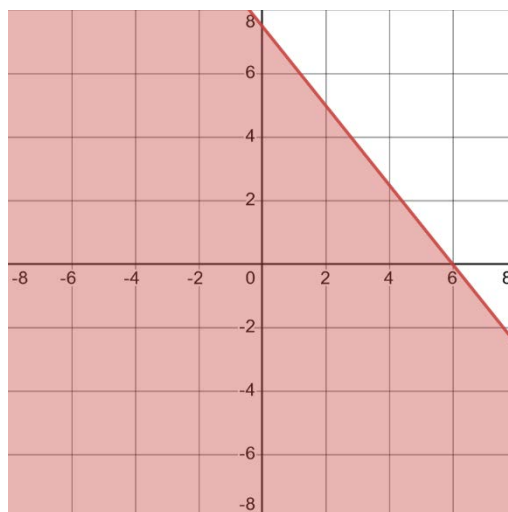
Let x represent lbs. of lunchmeat.

Let y represent lbs. of cheese.

2. Explain why the inequality sign $\leq$ makes sense in this context.

She cannot spend more than \$30, so it must be less than or equal to \$30.

3. Graph the inequality $5x + 4y \leq 30$.



4. The ordered pair $(5, -5)$ is a solution to the inequality $5x + 4y \leq 30$. Explain why this is not a viable solution in the context of buying lunchmeat and cheese.

You cannot buy a negative amount of anything.

5. The ordered pair $(2, 5)$ is a point on the boundary line. Explain what the ordered pair represents in the situation.

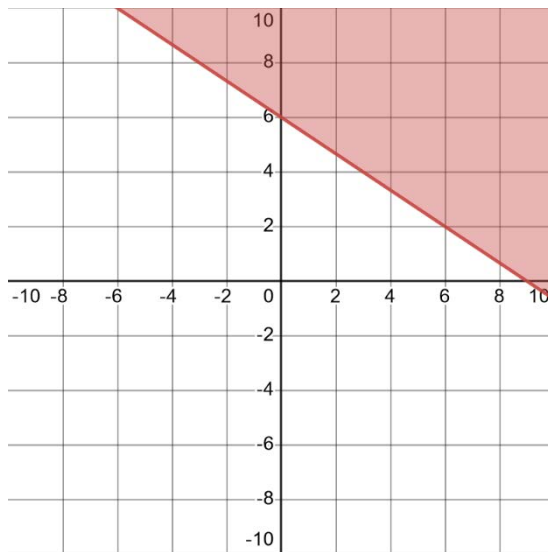
2 lbs of lunchmeat and 5 lbs of cheese can be bought for exactly \$30.

Talia invested some of her savings in the stock market. She bought 50 shares of a video streaming company and 75 shares of a manufacturing company. She hopes to make a combined profit of at least \$450 when she sells her shares. She wrote the inequality $50x + 75y \geq 450$ to represent the situation, where x is the change in value of one share of the streaming company and y is the change in value of one share of the manufacturing company.

6. Explain what it would mean if x or y were a negative number.

The value decreased.

7. Graph the inequality $50x + 75y \geq 450$.



8. Explain what the point $(14.75, -3.25)$ represents in Talia's situation.

The video streaming value increased by \$14.75 and the manufacturing company decreased by \$3.25.

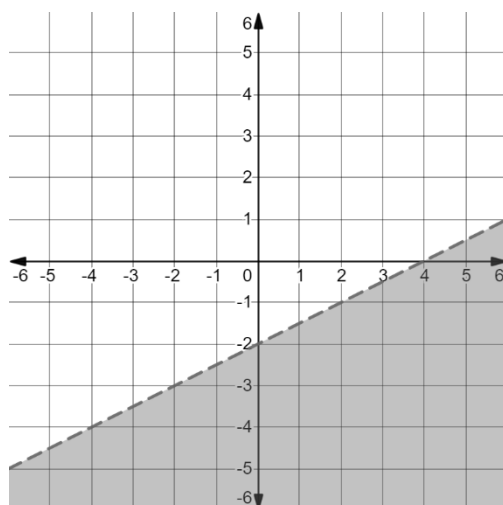
9. Explain why it makes sense in Talia's context that there are no solutions for which both x and y are both negative.

Because if both decreased, she can't have a profit.

10. Stock prices in the United States are rounded to the nearest hundredths place. Complete the statement:

In order for solutions to the mathematical inequality to be viable solutions in the real-world context, the x and y values must be rounded to 2 decimal places.

The graph of a *two* variable linear inequality is shown.



1. Write the inequality of the solution set graphed above in slope-intercept form.

$$y < \frac{1}{2}x - 2$$

2. Explain how to determine which inequality symbol to use when writing an inequality in slope-intercept form.

Since the line is dashed, the inequality sign will not be equal to. Since the graph is shaded below the line, select a point in the solution set and use the symbol that make it true.

3. Write the inequality of the solution set graphed above in point-slope form.

$$y + 1 < \frac{1}{2}(x - 2) \quad \text{Answers may vary – use any point on the dashed line.}$$

4. Explain how to determine which inequality symbol to use when writing an inequality in point-slope form.

Since the line is dashed, the inequality sign will not be equal to. Since the graph is shaded below the line, select a point in the solution set shaded region and use the symbol that makes the inequality true.

5. Write the inequality of the solution set graphed above in standard form.

$$y < \frac{1}{2}x - 2 \quad \rightarrow 2y < x - 4 \quad \rightarrow -x + 2y < -4 \quad x - 2y > 4$$

Five inequalities are given. Complete the table below to match each inequality to the graph of its solution set by writing the inequality on the line provided under each graph.

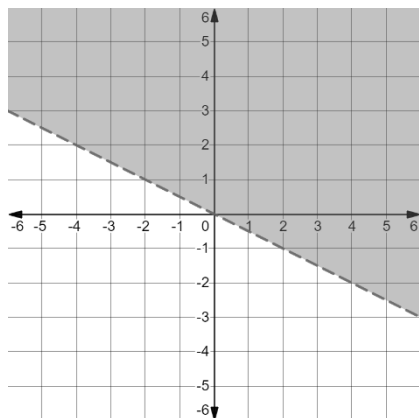
$$x \leq 4$$

$$y \leq -2x + 3$$

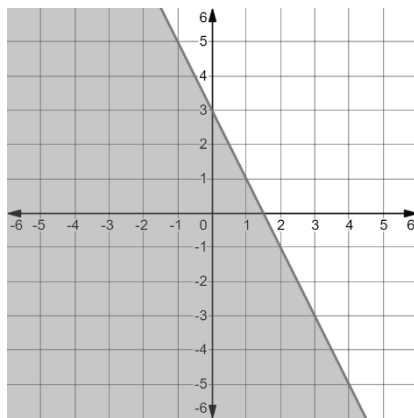
$$-x + 3y > -9$$

$$y > -3$$

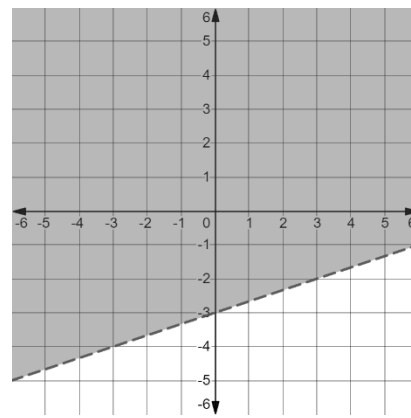
$$y + 2 > -\frac{1}{2}(x - 4)$$



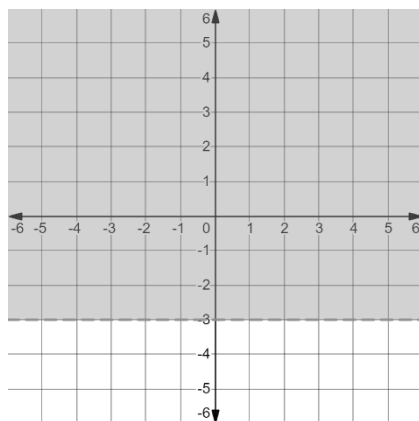
6. Inequality: $y + 2 > -\frac{1}{2}(x - 4)$



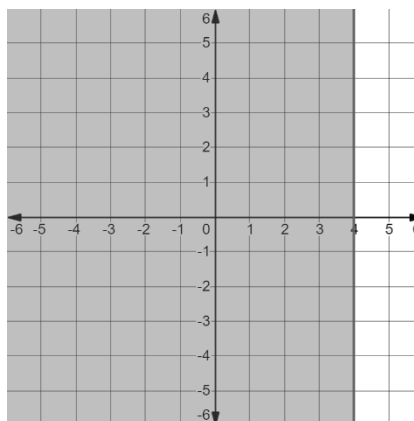
7. Inequality: $y \leq -2x + 3$



8. Inequality: $-x + 3y > -9$



9. Inequality: $y > -3$



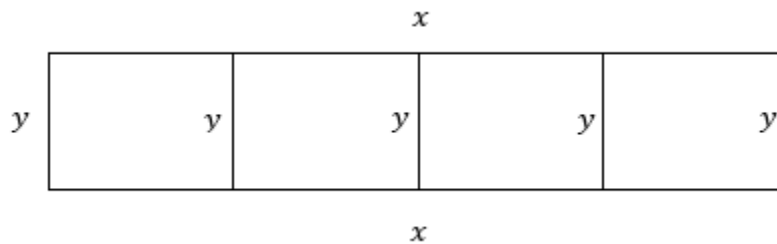
10. Inequality: $x \leq 4$

Felix wants to build a rectangular fence in his yard that will be divided into *four* equal pens for chickens. Each pen needs to be the same size. He has 30 feet of fencing to build the fence. This scenario can be modeled as a *two*-variable inequality.

1. Describe what the two variables would represent in the context of Felix's fence.

x will represent the length of the rectangle & *y* is the width of each section.

2. The diagram below shows a possible model of Felix's fence with sides labeled as *x* and *y*.



3. Recall Felix had 30 feet of fencing to use. Write a *two*-variable inequality that Felix could use to find possible dimensions for his fence.

$$2x + 5y \leq 30$$

4. Explain how to determine which inequality symbol to use.

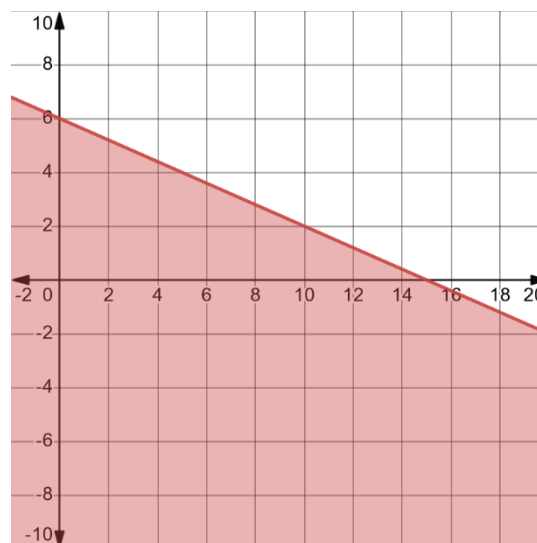
He only has 30 ft. so he can't use more than that.

5. Graph the inequality.

$$\begin{array}{rcl} 2x + 5y & \leq & 30 \\ -2x & & -2x \end{array}$$

$$\frac{5y}{5} \leq \frac{-2x}{5} + \frac{30}{5}$$

$$y \leq -\frac{2}{5}x + 6$$



Drew was applying for a personal loan to pay for home improvement projects. His bank requires a minimum of \$1,000 between his checking and savings accounts to be considered for a loan.

6. Define two variables for this situation.

Let x = checking account balance

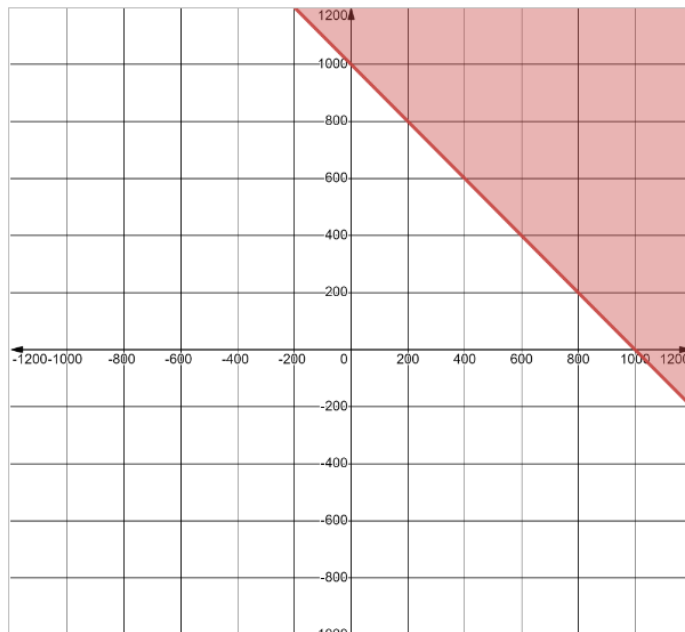
Let y = savings account balance

7. Write an inequality in terms of x and y that could be used to determine whether Drew meets the minimum requirement for the loan.

$$x + y \geq 1,000$$

8. Graph the inequality.

$$y \geq -x + 1000$$



9. Complete the statement.

To be a viable solution in this context, the x and y values must both be rounded to 2 decimal places.

10. Drew overdrew his checking account, which now has a balance of $-\$350.00$. He has $\$1,452.65$ in his savings account. Does he meet the minimum requirement?

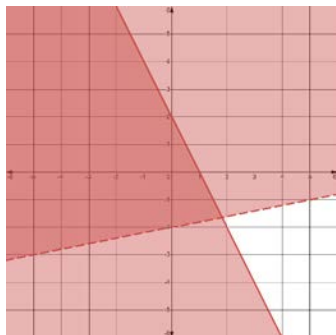
$$\begin{array}{r} 1452.65 \\ -350.00 \\ \hline 1102.65 \end{array}$$

Yes, he still has at least \$1,000.

17 < 4 17 is not < 4.

Consider the system of inequalities, $\begin{cases} y + 3 > 0.2(x + 5) \\ y - 4 \leq -2(x + 1) \end{cases}$ $\begin{cases} y > 0.2x - 2 \text{ or } y > \frac{1}{5}x - 2 \\ y \leq -2x + 2 \end{cases}$

6. Graph the boundary lines of the system.



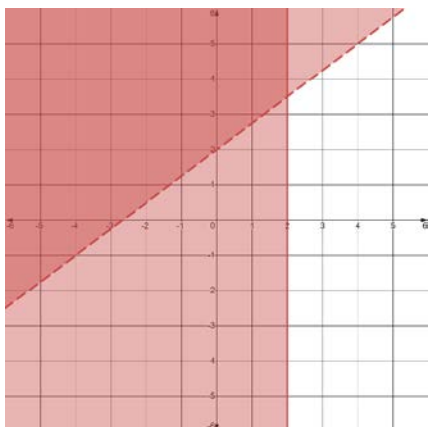
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<https://www.desmos.com/calculator/b7vfzjvv5>

Use substitution to determine whether the given ordered pairs are solutions to the system.

	$y + 3 > 0.2(x + 5)$	$y - 4 \leq -2(x + 1)$	Solution to the system?
7. $(-3, 4)$	$4 + 3 > 0.2(-3 + 5)$ $7 > 0.4 \checkmark$	$4 - 4 \leq -2(-3 + 1)$ $0 \leq 4 \checkmark$	☐ Yes ☐ No
8. $(5, 2)$	$2 + 3 > 0.2(5 + 5)$ $5 > 2 \checkmark$	$2 - 4 \leq -2(5 + 1)$ $-2 \leq -12 \times$	☐ Yes ☒ No

9. Complete the graph by shading the region included in the solution set of the system.

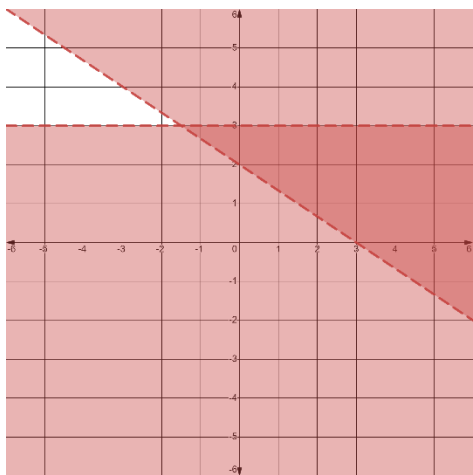
10. Graph the solution set of the system of inequalities: $\begin{cases} x \leq 2 \\ y > \frac{3}{4}x + 2 \end{cases}$



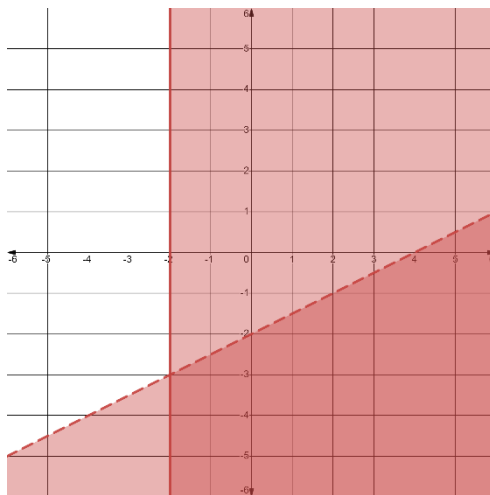
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Graph the solution set for each system of inequalities on the coordinate grid.

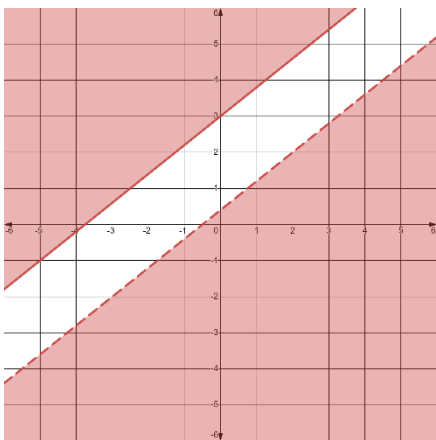
1. $\begin{cases} y < 3 \\ 2x + 3y > 6 \end{cases} \rightarrow y > -\frac{2}{3}x + 2$



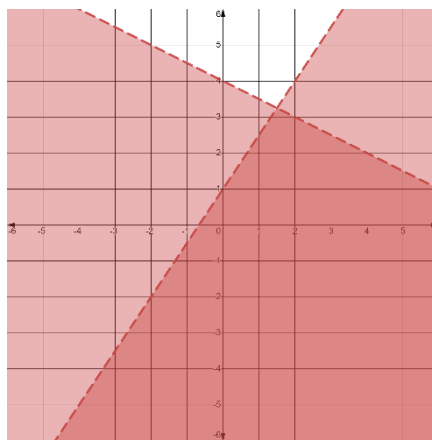
2. $\begin{cases} x \geq -2 \\ 2x - 4y > 8 \end{cases} \rightarrow y < \frac{1}{2}x - 2$



3. $\begin{cases} y \geq 0.8x + 3 \\ y - 2 < 0.8(x - 2) \end{cases} \rightarrow y \geq \frac{4}{5}x + 3$



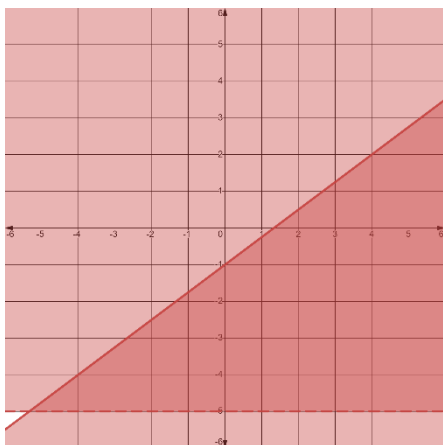
4. $\begin{cases} y < \frac{3}{2}x + 1 \\ y - 1 < -0.5(x - 6) \end{cases}$



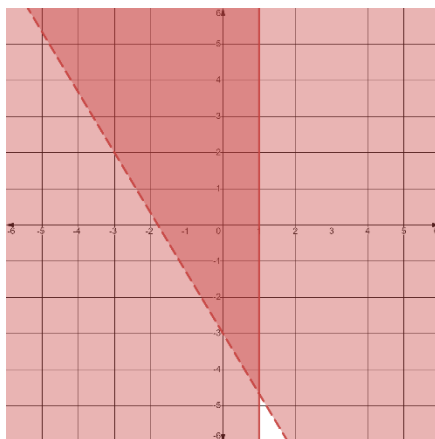
5. From problems 1-4, identify the graph that has no solution and explain your reasoning. **Answers may vary.**

#3 does not have a common shaded area, plus the slopes are the same which means they're parallel = no solution.

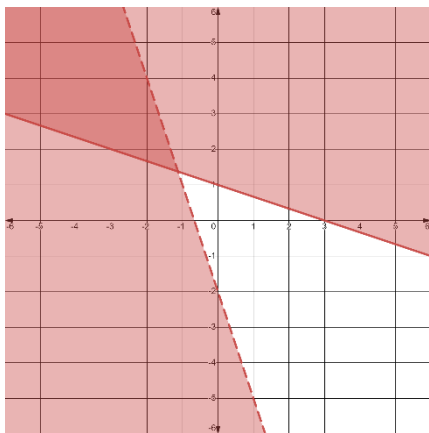
$$6. \quad \begin{cases} y > -5 \\ y + 4 \leq \frac{3}{4}(x + 4) \end{cases}$$



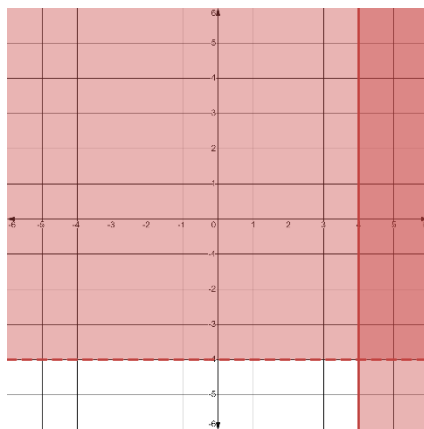
$$7. \quad \begin{cases} x \leq 1 \\ y > -\frac{5}{3}x - 3 \end{cases}$$



$$8. \quad \begin{cases} y \geq -\frac{1}{3}x + 1 \\ y < -3x - 2 \end{cases}$$



$$9. \quad \begin{cases} x \geq 4 \\ y > -4 \end{cases}$$



$$10. \quad \begin{cases} -4x - 2y \geq 10 \\ y \geq -\frac{5}{4}x - 3 \end{cases} \quad \rightarrow \quad y \leq -2x - 5$$

